

L2M

"Attention is all you need"

Review

"Food is delicious but the price is high"

Aspect $\rightarrow$ food $\rightarrow$ positive
 $\rightarrow$ Price $\rightarrow$ negative

```
INPUT = {  
  'text': 'food is delicious but price is high',  
  'aspect': 'food'  
}  
OUTPUT = {  
  'aspect_sentiment': 'Positive'  
}
```

```
✓ INPUT = {  
  'text': 'food is delicious but price is high',  
  'aspect': 'price'  
}  
✓ OUTPUT = {  
  'aspect_sentiment': 'Negative'  
}
```

Aspect based
Sentiment
Analysis
(ABSA)

Architecture Idea

- ↳ Tokenise review & aspect
- ↳ Embed $\rightarrow$ number (pre-trained TF Mechanism)
- ↳ concatenate [Token, Aspect] / Combining
- ↳ LSTM (Bi-directional LSTM)

→ Last State → Dense → Softmax → Positive / Negative

Review text ----> Embedding ----> LSTM ----> Dense ----> Sentiment
Aspect text ----> Embedding ----> (merged into the same pipeline)

Detailed Notes → 70-72% Accuracy

Is this enough?

→ LSTM → compress entire sequence into one single vector

→ long Reviews → Imp word (delicious, overpriced) may be far from the aspect

Food ✓ Price ✓

delicious

overpriced

Motivation → "Attention"

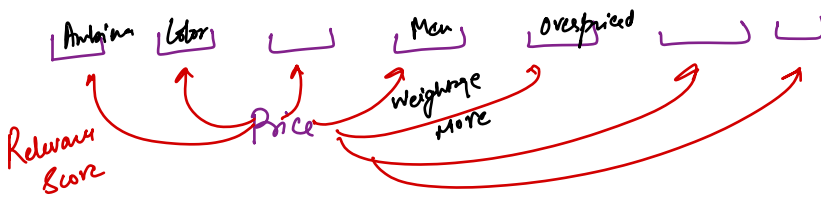
Trainable way to focus on specific part of sequence

Q How's the service

→ "Barber was rude" , "Service was bad"

Given

Similarity → Word embedding / hidden states
Query / Aspect



Attention → Relevance Score
 → Query / weights
 → Context Vector → weighted sum of the sequence

text: food is delicious but price is high
 weights: 0.10 0.05 0.35 0.05 0.30 0.05 0.10 (for aspect = food)

text: food is delicious but price is high
 weights: 0.05 0.05 0.10 0.05 0.30 0.10 0.35 (for aspect = price)

Mathematics → Attention Mechanism

- ① Key → Tokens
 - ② Query → Aspect
 - ③ Value
- ★ Energy
 - ★ Attention weight (α)
 - ★ Context Representation

Key

ABSA → Review words
 ↓
 Tokens → D dimension

$$K = \begin{bmatrix} \text{Food} \\ \text{way} \\ \text{delicious} \\ \text{price} \\ \text{way} \\ \text{high} \end{bmatrix} \in \mathbb{R}^{6 \times d} \quad d=3$$

6 tokens

$$K_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ "Food" } \quad K_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ "way" } \quad K_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ "delicious" } \quad K_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ "price" } \quad K_5 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \text{ "way high" }$$

$K_6^0 \quad K_7^0$

Embedding → one hot encoded

✓ → OpenAI embedding

✓ → word2vec

✓ → Bert

✓ → GPT-3 embedding

TPIDF

Query → "what we are looking for"

↳ "price"

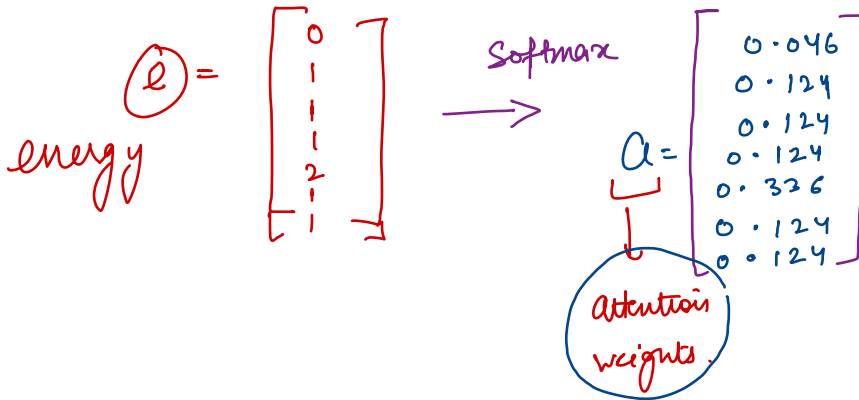
$$Q = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Energy / Alignment function

$$Q^T \cdot K$$

- $e_1 = Q^T K_1 = [0, 1, 1] \cdot [1, 0, 0] = 0$
- $e_2 = [0, 1, 1] \cdot [0, 1, 0] = 1$
- $e_3 = [0, 1, 1] \cdot [1, 1, 0] = 1$
- $e_4 = [0, 1, 1] \cdot [0, 0, 1] = 1$
- $e_5 = [0, 1, 1] \cdot [0, 1, 1] = 2$
- $e_6 = [0, 1, 1] \cdot [0, 1, 0] = 1$
- $e_7 = [0, 1, 1] \cdot [1, 0, 1] = 1$

token '5' → Price
Max score = 2

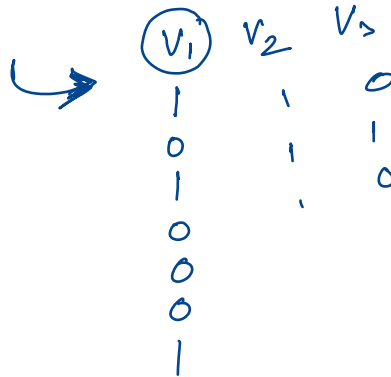
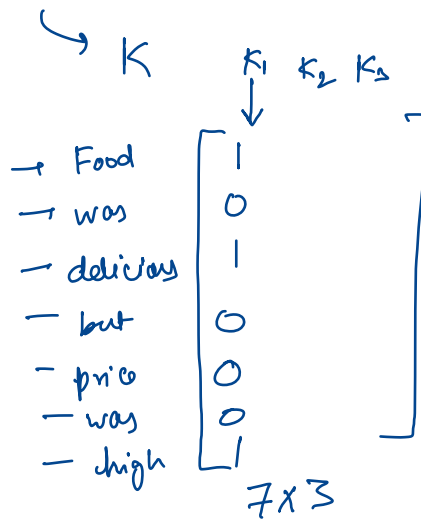


Generally, weight are multiplied with values

Values

$V \Rightarrow K$
shape ↓ shape
7x3 ↓ 7x3

7 Rows 3 cols
7 words 3 dim



Context Vector

$$C_1 = a \times V_1$$

$$C_2 = a \times V_2$$

$$C_3 = a \times V_3$$

$$C_1 = 0.046 \cdot 1 + 0.124 \cdot 0 + 0.124 \cdot 1 + 0.124 \cdot 0 + 0.336 \cdot 0 + 0.124 \cdot 0 + 0.124 \cdot 1$$

$$\approx 0.046 + 0 + 0.124 + 0 + 0 + 0 + 0.124 = 0.294$$

$$C_2 \checkmark$$

$$C_3 \checkmark$$

C is a single vector $\rightarrow$ Summarises the Review
 $\rightarrow$ Also biased towards tokens relevant to the Query.

Energy

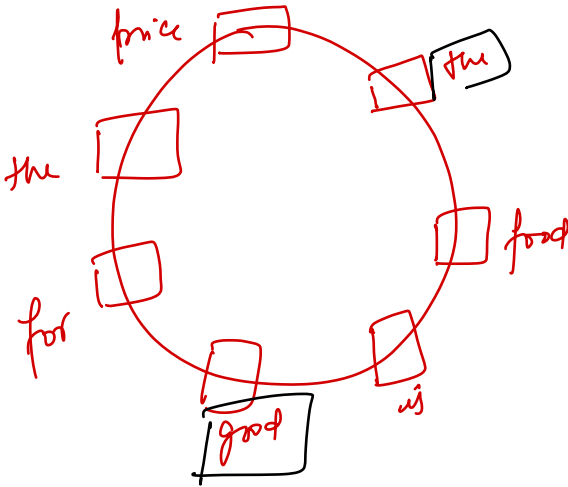
- ① Dot Product
- ② Cosine Similarity
- ③ Scaled Dot Product

Notes for Details

Attention Mechanism

① Self Attention

Key = Query = Value



The cat sat on the mat. It was dirty.

→ Self Attention / Multi-headed attention

Post-read (Soft) / Hard Attention

✓ Head 1 → Grammar head → Verb → noun

✓ Head 2 → Sentiment head → great | bad | cheap

✓ Head 3 → Aspect head